

Dheeraj Karanam

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EDUCATION

Indiana University Bloomington

May 2027

M.S. in Data Science – Luddy School of Informatics, Computing, and Engineering | GPA: 3.94/4.0

Andhra University

2019

B.E. in Electronics & Communication Engineering

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, FastAPI, pytest), SQL, R

LLMs & GenAI: LangChain, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Vector Databases (FAISS), GenAI Applications & Agents, Prompt Engineering, LLM APIs (Anthropic, OpenAI, HuggingFace), LoRA

Machine Learning: Scikit-learn, PyTorch, XGBoost, Statistical Analysis, Feature Engineering, Model Evaluation, SHAP

Software & MLOps: Git, CI/CD (Jenkins), REST APIs & Microservices, Docker, Airflow, MLflow, Databricks, Apache Spark / PySpark, Agile / Jira

Cloud & Data: AWS (Lambda, Bedrock, IAM, DynamoDB), GCP, BigQuery, PostgreSQL, Hadoop, Hive

Analytics & Visualization: Business Analytics, Data Quality & Validation, Tableau, Power BI, Matplotlib, Seaborn

Domain: Financial Services (Retail Banking, Risk & Portfolio Analysis)

WORK EXPERIENCE

McAfee

Frisco, TX

AI Intern

May 2026 – Aug 2026

Platform Engineering – AWS AI Frontier Agents Integration

- Built a **GenAI agent** pipeline routing **Grafana** alerts through **AWS Lambda** into **Bedrock AgentCore**, automating first-pass incident triage and root cause analysis for Platform Engineering and SRE 24/7 teams.
- Integrated **MCP** servers into the agent stack on AWS, exposing internal tooling and runbook actions to Bedrock agents as callable tools; provisioned Lambda functions, IAM roles, and DynamoDB tables as code with Terraform for repeatable deployment.
- Implemented cross-account **IAM** role assumption and **DynamoDB**-based deduplication so repeated alerts resolve to a single agent invocation; enforced **OPA** policy checks on agent-proposed actions to keep automation inside approved guardrails.
- Integrated commit-triggered AI code review into the pre-deployment path, surfacing security and code quality issues prior to merge; instrumented telemetry on agent usage to measure adoption and drive continuous prompt tuning.

OCTO – Text Scam Detector Explainability

- Built a **SHAP**-based token-wise explanation module for an LLM scam-detection classifier, surfacing which spans of a message drove each prediction; implemented embedding-based **top-k nearest neighbor** retrieval to ground predictions in similar training examples.
- Prototyped a **Databricks**-hosted application unifying multiple explanation methods for side-by-side model comparison; documented methodology and presented findings and recommendations to the team.

IBM – Barclays Retail Banking

Hyderabad, India

Data Engineer

Mar 2023 – Aug 2025

- Engineered and optimized batch data pipelines on **Hadoop/Spark** with **Ab Initio** processing **10M+ rows** of retail banking data; built scalable data models with end-to-end data lineage for downstream analytics and ML processes consumed by 5+ teams.
- Designed and deployed automated data validation and quality frameworks (schema drift, null checks, outlier detection, row-count reconciliation) in Python and SQL with unit test coverage; reduced debugging effort by **80%**.
- Led **platform migration** of 108 workflows and 448 ETL jobs (Cloudera CDH to BDH, RHEL6 to RHEL8) using CI/CD pipelines with Jenkins; improved system performance by **25%**; established data contracts and schema evolution standards across 3 business units.
- Collaborated cross-functionally in Agile delivery on a global banking client account, presenting pipeline architecture and data quality findings to both technical engineering teams and non-technical client stakeholders; trained and guided 6 junior engineers.

TCS – Virgin Media

Chennai, India

Associate Software Engineer

Jul 2019 – Feb 2023

- Trained and deployed an **NLP classification model** (Rasa NLU/Core) to production at **90% F1 score**, serving it via REST API on **AWS** with unit test coverage and post-deployment monitoring; presented results at a client town hall, driving adoption across 2 additional teams.
- Architected **SCD Type 1/2** temporal data models for Netezza-to-GCP BigQuery migration across **645 tables**; applied schema evolution principles ensuring backward compatibility and data integrity during production cutover; delivered with Jenkins-based deployment automation.
- Optimized **Spark SQL** transformations (partition pruning, predicate pushdown, partitioning and clustering rewrites) compressing a 14-hour pipeline to 12 hours; documented execution plan analysis and bottleneck findings for the engineering team.
- Maintained **99% SLA** and 99.5% uptime across batch and near-real-time pipelines; recovered 50+ failed critical jobs in Control-M by implementing error-handling and monitoring protocols ensuring business continuity.
- Built **Tableau and Power BI** dashboards with interactive visualizations (filters, slicers, KPI measures) translating complex pipeline metrics and data quality findings into actionable insights for non-technical business leadership.

PROJECTS

Financial Insight System – RAG, LoRA, MCP | *finance_rag_lora_mcp*

- Built end-to-end data ingestion and curation pipeline processing **500+ financial documents** (PDF, JSON unstructured data) with a **vector database** for retrieval; implemented ML-based filtering, schema-aligned extraction, confidence scoring, and data quality validation for LLM consumption.
- Orchestrated multi-step **agentic workflows** through **MCP** tool calling; built a QA dashboard (**FastAPI**) tracking output quality (BERTScore, F1), model latency, and data drift; delivered with unit-tested code, Git version control, and runbook-style documentation.

CAG vs RAG Evaluation Framework – Medical Genomics | *Prof. Vardhan, IU*

- Built an **LLM-based evaluation framework** (LangChain, FAISS vector store) benchmarking **RAG, CAG, and CRAG** on a biomedical question-answering benchmark across Llama 3.1, OpenBioLLM, and BioMistral; RAG outperformed both CAG and CRAG on **BERTScore** and **Token F1**.
- Profiled all three engines on **BigRed200 HPC** (SLURM, A100 GPUs); characterized the **latency vs accuracy trade-off**, with CAG substantially reducing latency at the cost of lower Token F1; delivered a reproducible benchmarking framework with consistent data quality controls.

Modular RAG Pipeline – Accounting Document Extraction | *Prof. Golshan, IU*

- Building a **modular RAG pipeline** that ingests long accounting and financial filings (PDF, text), performs **context-aware chunking**, retrieval, and aggregation within model context limits, and emits **structured JSON/tabular** output for downstream analysis.
- Architecting the system around **swappable components** (input source, prompt template, output schema) so pipelines reconfigure without touching core code; running experiments on **IU BigRed200 HPC** (SLURM, A100 GPUs).

CERTIFICATIONS

MITx ML with Python (2024) | MITx Statistics (2024) | MITx Probability (2024) | IBM Python for DS (2023)